

Notes on Polynomial Regression

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DEFINITION

Polynomial regression is a type of regression analysis in which the relationship between the independent variable x and the dependent variable y is modeled as an n -th degree polynomial. The general form of a polynomial regression model is:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_n x^n + \epsilon$$

where $\beta_0, \beta_1, \dots, \beta_n$ are the coefficients of the polynomial, and ϵ is the error term.

DIFFERENCE BETWEEN LINEAR REGRESSION AND POLYNOMIAL REGRESSION

Linear regression models the relationship between variables as a straight line, while polynomial regression models the relationship as a curve of a specified degree.

WHEN TO USE POLYNOMIAL REGRESSION

Polynomial regression is useful when the relationship between the independent and dependent variables is non-linear. It can capture more complex patterns in the data compared to linear regression.